Network Security and Application Security Engineering Roadmap

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Comprehensive roadmap covering all essential domains required to become a Network and Application Security Engineer.

1. Core Foundations

- OSI Model, TCP/IP, Ports and Protocols
- Linux Basics: File system, permissions, processes
- Windows basics
- Security fundamentals: CIA Triad, threats, vulnerabilities, risk
- Basic attacks: DoS, phishing, malware

2. Networking Deep Knowledge

- Subnetting and IP addressing
- DNS, DHCP, ARP
- TCP vs UDP, packet structure
- Routing basics (OSPF, BGP), VLANs, NAT, load balancing

3. Network Security Engineering

- Firewalls, IDS, IPS, VPN
- Network segmentation and Zero Trust
- SIEM basics and monitoring
- Tools: Wireshark, Nmap, Snort

4. Application Security

- HTTP/HTTPS protocol, sessions and cookies
- OWASP Top 10 vulnerabilities
- SQL Injection, XSS, CSRF, IDOR
- Burp Suite and web testing tools

5. Secure Coding

- Input validation and sanitization
- Authentication and authorization
- Session management
- API security and secure development practices

6. System Security

- Linux security and permissions
- Process and memory basics
- Privilege escalation
- System hardening

7. Monitoring and Logging

- Log analysis and detection
- tcpdump, netstat, ss tools
- Security monitoring workflows

8. Offensive Security

- Penetration testing basics
- Reconnaissance and exploitation
- CTFs and practice labs
- Platforms: TryHackMe, Hack The Box

9. Real Projects

- Packet Sniffer
- Network Scanner
- Vulnerability Scanner
- Secure Authentication System
- Log Analyzer Dashboard

10. Certifications and Practice

- CCNA, Security+, CEH, eJPT
- Hands-on labs and continuous practice